

From Benchmark to Better Embeddings: Leveraging Synonym Substitution to Enhance Multimodal Models in Ukrainian

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Abstract

Multimodal retrieval models, such as Open-CLIP, rely on aligned textual and visual representations, yet their robustness to lexical variations remains underexplored, especially in low-resource languages. In this study, we introduce three methods for generating synonym substitutions in Ukrainian: a dictionary-based approach, a large language model (LLM)-generated approach, and a hybrid method combining both. We further demonstrate that fine-tuning Open-CLIP with synonym-augmented data improves retrieval robustness, leading to a 7% increase in HIT@5. Our findings provide insights into synonym substitution techniques for low-resource languages and offer a pathway to enhancing the robustness of multimodal models in diverse linguistic settings.

1 Introduction

Multimodal retrieval systems, such as those based on CLIP (Radford et al., 2021), rely on aligning textual and visual representations. An open problem in these systems is ensuring that small linguistic variations—such as synonym substitutions—do not lead to markedly different retrieval outcomes. In an ideal scenario, substituting a word with a synonym would yield the same or similar images; however, biases in training data can cause the model to favor specific synonyms, resulting in inconsistent performance. This issue is particularly concerning in low-resource languages, where models have limited exposure to diverse synonymous expressions during training.

Recent studies have examined the adversarial robustness of multimodal models. For instance, (Zhou et al., 2024) demonstrated that adversarial perturbations in both text and image modalities can degrade performance. Similarly, (Liu, 2024) compared multimodal and unimodal models under text-based attacks and found that, despite some improvements, multimodal models remain vulnerable to lexical

variations. Notably, much of this research has focused on high-resource languages, leaving the robustness of these models in low-resource language contexts relatively underexplored.

In this study, we evaluate the robustness of multilingual multimodal models for retrieval tasks in a low-resource language setting using the Open-CLIP (Cherti et al., 2023) model and a Ukrainian version of the Multi30K dataset (Saichyshyna et al., 2023). Our contributions are threefold: (i) we establish a baseline for robustness evaluation under synonym substitutions for Ukrainian language, (ii) we conduct a detailed analysis of model performance under lexical variations using three synonym substitution pipelines—a dictionary-based approach, a large language model approach and hybrid, and (iii) we demonstrate that fine-tuning with synthetic synonym replacements enhances robustness, as evidenced by an average ~7% increase in HIT@5 after synonym substitution.

2 Related Work

Multimodal text-to-image models mostly follow the CLIP-like approach (Radford et al., 2021), which enables joint learning of image and text representations. While the original CLIP model was trained monolingually, several successful adaptations have extended its capabilities to multilingual settings. Notable examples include mCLIP (Chen et al., 2023) and Cross-lingual CLIP (Carlsson et al., 2022), which employ a teacher-student transfer learning approach to adapt CLIP to multiple languages. Another strategy involves replacing the CLIP text encoder with a multilingual language model and training it on a multilingual dataset, as demonstrated in Open-CLIP (Cherti et al., 2023). We evaluated both approaches on the Ukrainian version of Multi30K and found that Open-CLIP outperformed the teacher-student methods, making it the primary model for our study.

A widely used method for assessing model robustness involves adversarial attacks, which introduce perturbations to test a model’s sensitivity to input variations. In this work, we focus on text-based robustness, specifically in the Ukrainian language, by evaluating the model’s sensitivity to synonym substitutions. Synonym-based adversarial attacks are well-studied in high-resource languages such as English, where various approaches exist to generate semantically and syntactically valid substitutions (Jin et al., 2020; Li et al., 2020). However, these methods rely on lexical resources like WordNet or pretrained Counter-fitting embeddings (Mrkšić et al., 2016), which are not available for low-resource languages like Ukrainian. Furthermore, many synonym substitution techniques have been found to generate invalid replacements, as highlighted in (Chiang and Lee, 2023).

Given these limitations, instead of directly adapting existing synonym substitution techniques from English, we propose a novel approach leveraging large language models (LLMs) to generate contextually and grammatically appropriate synonym replacements for Ukrainian.

3 Methodology

3.1 Problem Defenition

We define our primary objective as evaluating and improving the robustness of multimodal embeddings, specifically Open-CLIP, to synonym substitutions in a low-resource, morphologically rich language setting. Our experiments focus on Ukrainian, using Multi30K as the underlying dataset.

First, we establish baseline performance by measuring how well Open-CLIP retrieves images using the original English and Ukrainian captions from Multi30K utilizing the key retrieval metrics—HIT@1, HIT@5, and Mean Reciprocal Rank (MRR)—for both English and Ukrainian test sets. As shown in Table 1, the model achieves a HIT@1 of 0.416 and a HIT@5 of 0.657 for English, but these metrics drop to 0.321 and 0.543 for Ukrainian, respectively. These results indicate a performance gap between English and Ukrainian.

To assess robustness, we measure how much retrieval accuracy degrades when we apply synonym substitutions to the Ukrainian captions. Unlike English, where WordNet and other lexical resources facilitate systematic synonym replacements, Ukrainian’s morphological complexity complicates this process: valid synonyms must match

the grammatical form of the original word to retain meaning. In the following Approach section 3.2, we describe our method for systematically generating Ukrainian synonyms and evaluating their impact on image retrieval performance.

To assess robustness, we measure how much retrieval accuracy degrades when we apply synonym substitutions to the Ukrainian captions. To ensure a clear experimental setup and isolate the effect of synonym substitutions, we evaluate robustness only on a subset of Multi30K where the original model correctly retrieves the image. This approach helps mitigate potential errors from the model or dataset itself, allowing us to focus specifically on the impact of lexical variations. Unlike English, where WordNet and other lexical resources facilitate systematic synonym replacements, Ukrainian’s morphological complexity complicates this process: valid synonyms must match the grammatical form of the original word to retain meaning. In the following Approach section 3.2, we describe our method for systematically generating Ukrainian synonyms and evaluating their impact on image retrieval performance.

Language	HIT@1	HIT@5	MRR
English	0.416	0.657	0.527
Ukrainian	0.321	0.543	0.426

Table 1: Baseline performance of Open-CLIP on English and Ukrainian Multi30K test sets (1,941 samples).

3.2 Approach

Generating high-quality synonym substitutions for Ukrainian presents challenges due to its rich morphology and inflectional complexity. Unlike English, where words generally appear in a single invariant form, Ukrainian words change with number, gender, and case, making it difficult to preserve semantic equivalence during automated substitution. Furthermore, the lack of modern synonym substitution frameworks exacerbates the challenge of evaluating model robustness.

To address these issues, we explore three synonym generation approaches. The first method is dictionary-based, using the Словник синонімів української мови (Наукова думка, 1999), which contains 9,200 synonym groups. Although this resource offers broad lexical coverage, it is prone to parsing errors, outdated terms, and domain-specific constraints that can result in grammatical mismatches; therefore, after selecting a

synonym from the dictionary, we apply Pymorphy to adjust its form to fit the sentence context.

The second method leverages a large language model (GPT-4o (OpenAI, 2024)) to generate context-aware synonyms that are both semantically and grammatically appropriate.

Recognizing the strengths and limitations of both methods, we propose a third method, the hybrid approach in which dictionary candidates are refined by the language model to select synonyms that best maintain sentence meaning and grammatical integrity.

Synonym Substitution Method	HIT@1	HIT@5	MRR
Dictionary	0.206	0.512	0.358
GPT-4o	0.335	0.676	0.49
Hybrid	0.446	0.776	0.593

Table 2: Performance of Open-CLIP on Ukrainian Multi30K test set by different synonym substitution methods.

Table 2 presents the image retrieval performance of Open-CLIP under three different synonym-substitution strategies. It shows that the choice of synonym substitution strategy has a substantial impact on retrieval performance. A purely dictionary-based approach yields the lowest scores across all metrics. We attribute this drop in accuracy to the dictionary’s limited contextual awareness: while it offers broad lexical coverage, the synonyms it provides often fail to match the morphological and semantic context of each sentence.

In contrast, the GPT-based approach significantly improves performance by generating synonyms that better preserve both meaning and grammatical form, thereby reducing unintended shifts in the text representation.

The best results, however, come from the hybrid method, in which dictionary candidates are refined by a large language model. This combined strategy benefits from the dictionary’s lexical breadth and the LLM’s context sensitivity, leading to more precise synonym matches and higher overall retrieval scores.

Despite these gains, further improvements remain necessary, particularly for morphologically rich and low-resource languages like Ukrainian.

3.3 Robustness Enhancement

To investigate whether the observed synonym substitution issues are primarily attributable to insufficient training data, we first fine-tune Open-CLIP on the Ukrainian portion of the Multi30K training set. This baseline fine-tuning approach allows us to assess whether additional in-language data alone can narrow the performance gap.

However, given the hypothesis that many multilingual models are trained on machine-translated data—and thus may inherit biases from automated translation systems—we further expand our training corpus by incorporating sentences with synonym substitutions generated through the GPT-based method. By enriching the training set with diverse lexical variants, we aim to strengthen the model’s robustness to subtle linguistic shifts commonly overlooked in low-resource languages.

We keep most Open-CLIP layers frozen to retain general multilingual knowledge, while selectively unfreezing a subset of text and vision layers for targeted adjustments. The precise hyperparameters are detailed in Appendix B.

4 Results

Table 3 demonstrates that fine-tuning improves performance across all synonym substitution methods. For example, under the hybrid approach, HIT@5 increases from 0.776 in the original model to 0.816 after Multi30k fine-tuning, and further to 0.839 with synonym-augmented fine-tuning. This pattern holds across methods: while standard fine-tuning yields noticeable gains, incorporating GPT-generated synonyms during fine-tuning consistently results in better performance, particularly as measured by HIT@5.

Although fine-tuning on the Multi30K dataset alone does enhance robustness by giving the model more extensive exposure to Ukrainian text, the most substantial gains emerge when we expand the training corpus with diverse synonym substitutions. A likely reason is that many less common or informal Ukrainian terms are rare or absent in the original pretraining data. As a result, the model sometimes misaligns text and images when encountering these unfamiliar synonyms. In addition, the model can develop biases associating certain synonyms with specific contexts, causing it to fail when these words appear alongside contradictory descriptors. By systematically introducing varied synonyms—particularly those generated or

Method	Model	HIT@1	HIT@5	MRR
Dict	Original Open-CLIP	0.206	0.512	0.358
	Multi30k Fine-Tuned	0.271	0.572	0.41
	Synonym-Augmented Fine-Tuned	0.261	0.594	0.414
GPT	Original Open-CLIP	0.335	0.676	0.487
	Multi30k Fine-Tuned	0.403	0.733	0.548
	Synonym-Augmented Fine-Tuned	0.443	0.757	0.585
Hybrid	Original Open-CLIP	0.446	0.776	0.593
	Multi30k Fine-Tuned	0.519	0.816	0.651
	Synonym-Augmented Fine-Tuned	0.54	0.839	0.669

Table 3: Test results for three synonym substitution methods (Dict, GPT, Hybrid) with three different models (Original Open-CLIP, Multi30k Fine-Tuned, Synonym-Augmented Fine-Tuned).

refined by GPT—the model learns a broader lexical range and gains exposure to a wider spectrum of linguistic contexts.

The phenomenon can be illustrated with the example caption "Чоловік у синій сорочці лагодить велосипед у жовтій кімнаті" (A man in a blue shirt is fixing a bicycle in a yellow room). Using attention-based prediction visualization technique proposed by (Chefer et al., 2021), we observe that Open-CLIP correctly aligns the token "велосипед" (bicycle) with the corresponding image region, but fails when the synonym "ровер" is used. In this case, the tokenizer splits "ровер" into "ро" and "вер", with the latter receiving no importance, thereby reducing text-image alignment accuracy. This suggests that the model not only lacks exposure to certain synonyms but also exhibits biases toward more frequently occurring terms in its pretraining data. See Appendix A

To evaluate synonym substitution quality, we manually reviewed 500 replacements per method, categorizing them as (1) correct, where the synonym fits both meaning and grammar; (2) correct word but grammatically incorrect; and (3) incorrect, where the meaning is altered or lost. Results are presented in Tabel 4.

5 Conclusion

In conclusion, our study underscores a critical vulnerability in multimodal embeddings: their sensitivity to subtle lexical variations, particularly in low-resource languages like Ukrainian. By systematically evaluating the effects of synonym substitutions through dictionary-based, GPT-4o, and hybrid approaches, we demonstrated that minor linguistic shifts can significantly degrade image

Replacement Method	Dict	GPT	Hybrid
Correct	35.2	84.6	90.4
Grammar	27.6	2.4	1.2
Semantic	37.2	13	8.4

Table 4: Manual evaluation of synonym substitution quality across different methods, with results presented as percentages.

retrieval performance. Our solution, fine-tuning with synonym-augmented data, improves robustness by reducing the model’s sensitivity to lexical variations and ensuring more consistent retrieval performance. To build on this foundation, future research should explore more adaptive and balanced training strategies.

Limitations

While our study provides insights into the robustness of multimodal models under synonym substitution in a low-resource language, several limitations remain. First, our synonym substitution strategies rely on predefined lexical resources and large language models, which may introduce biases based on their training data. The dictionary-based method lacks contextual awareness, while the LLM-based approach depends on the quality of its training corpus and may generate synonyms that do not always preserve the intended meaning. Second, our evaluation is limited to the Ukrainian language using the Multi30K dataset, which, although a valuable benchmark, may not fully represent the linguistic diversity and complexity of real-world multimodal retrieval tasks.

Ethical Considerations

This study focuses on enhancing the robustness of multimodal retrieval models and does not involve human subjects, personal data, or sensitive information. However, ethical concerns arise regarding potential biases in the synonym substitution methods and the pretraining data of the models used. The dictionary-based approach may contain outdated or culturally specific synonyms that do not reflect contemporary language use. Additionally, large language models, such as GPT-4o, can inherit biases from their training data, which may influence synonym generation and, consequently, model predictions.

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A Attention-based Prediction Visualization

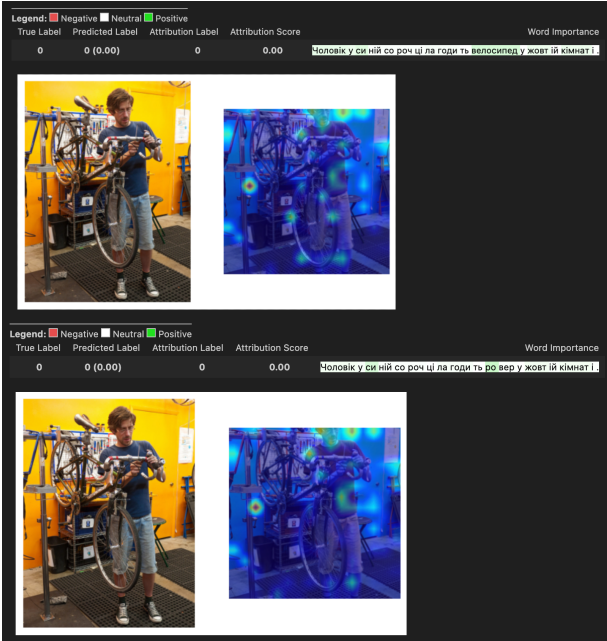


Figure 1: Attribution maps for Open-CLIP image-text alignment. The upper example uses "велосипед" (bicycle), which the model correctly aligns with the image. The lower example replaces it with the synonym "ровер", causing the tokenizer to split it into subwords ("po" and "вер"), with "вер" receiving no importance, reducing retrieval accuracy.

B Hyperparameters

The Open-CLIP model used in our experiments is based on the ViT-H/14 architecture, which contains approximately 632 million parameters. We conducted both fine-tuning on the Multi30K dataset and fine-tuning with synonym-augmented data on a single NVIDIA RTX 3090 GPU, each taking approximately 10 hours to complete. This setup allowed us to efficiently adapt Open-CLIP for improved robustness while maintaining a manageable computational cost.

Table 5 lists all hyperparameters used in our experiments.

C Dataset and Model Details

All the artifacts used in this study are open-source and available for research purposes. The Multi30K dataset (Ukrainian version) is publicly accessible and follows an open-source research license. The Open-CLIP model is released under an Apache 2.0 license, allowing for modification and redistribu-

tion within the scope of open research. Any synonym substitution dataset we generate will be made available under a Creative Commons (CC-BY) license to ensure transparency and reproducibility while complying with open-access principles.

Parameter	Value
-lr	1×10^{-6}
-warmup	1000
-batch-size	64
-epochs	20
-accum-freq	2
-wd	0.01
-lock-image	True
-lock-image-unlocked-groups	2
-lock-text	True
-lock-text-unlocked-layers	6
-pretrained	frozen_laion5b_s13b_b90k
-model	xlm-roberta-large-ViT-H-14

Table 5: Hyperparameters used in the experiment.